



Published in final edited form as:

Arch Dis Child. 2020 July ; 105(7): 631–638. doi:10.1136/archdischild-2019-317652.

Evidence of the non-linear nature of skeletal maturation

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Abstract

Objective—The aim of this study was to assess longitudinal trajectories of skeletal maturation to determine if children exhibit periods of rapid maturation during normal childhood and adolescence.

Design—Retrospective longitudinal study. Patients: 345 participants, with an average of 25 assessments per participant, between 3 and 20 years of age from the Fels Longitudinal Study.

Main outcome measures—Chronological age (ie, timing) and rate (ie, tempo) of skeletal maturation, as assessed by the Fels Method, at each maturational milestone, as well as the duration of time spent between any two milestones, were calculated for each participant-specific maturational trajectory and compared between three unique, non-linear maturational trajectory types.

Results—More than 81% of participants exhibited a rapid period of skeletal maturation during childhood and/or adolescence, most of whom were characterised by a single maturational spurt during adolescence. Participants with only a single adolescent spurt in skeletal maturation reach adolescent onset and peak approximately 2.8 and 4.2 years earlier, respectively, in boys ($p < 0.001$) and girls ($p < 0.001$), than when compared with participants with both childhood and adolescent spurts. Differences in the timing and tempo of maturational milestones were driven primarily by trajectory type.

Conclusions—Rapid changes in skeletal maturation occur during normal childhood and/or adolescence, indicating the presence of a maturational spurt: a developmental phenomenon that has remained largely uncharacterised. This work highlights patterned changes in the timing, tempo

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Contributors MEB: conceptualisation, investigation, methodology, visualisation, formal analysis, writing (original draft), review and editing. EVL: data curation, methodology, visualisation, formal analysis, writing, review and editing. RJS: resources, methodology, writing, review and editing. DLD: conceptualisation, resources, investigation, data curation, methodology, visualisation, writing, review and editing. All authors approved the final version of the manuscript.

Competing interests None declared.

Patient consent for publication Not required.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request according to regulations of the National Institutes of Health and the Fels Longitudinal Study. Requests should be sent to Dr Dana L Duren (durendl@health.missouri.edu).

and duration of longitudinal skeletal maturation while simultaneously shifting the paradigm that skeletal maturation progresses linearly.

INTRODUCTION

A detailed assessment of a child's skeletal maturity provides critical insight related to their biological maturity status, including their relationship to other growth and/or developmental milestones (eg, peak height velocity). Single, cross-sectional skeletal maturity assessments are expected to track linearly with chronological age throughout childhood and adolescence¹²; however, individual variability in the timing and tempo of longitudinal skeletal maturation can lead to deviations in the linear paradigm, clouding interpretations, particularly those related to the diagnosis, treatment and management of many skeletal growth and/or developmental disorders.^{3–8} Fluctuations in the tempo of skeletal maturation throughout childhood and adolescence have been previously characterised^{29–12}; however, participant-specific differences in the timing, tempo and duration of these fluctuations as they relate to maturational milestones (eg, adolescent peak) have yet to be fully elucidated and applied to interpretations of maturational deviations identified during cross-sectional skeletal maturity assessments.

The aim of the present study was to identify and characterise changes in the timing, tempo and duration of individual skeletal maturational trajectories throughout childhood and adolescence. We hypothesise that (1) most children will exhibit a maturational spurt, akin to that of the adolescent growth spurt, characterised by normal changes in the timing and tempo of skeletal maturation, and (2) that the timing, tempo and duration of maturational changes are related to the overall maturational trajectory.

MATERIALS AND METHODS

Study sample

The study sample includes participants from the Fels Longitudinal Study (FLS), all of whom are from southwest Ohio and of primarily European descent (<1% other). Participants were assessed beginning at birth, every 3 months for the first year of life and every 6 months afterward until reaching adulthood.¹³¹⁴ Inclusion in the present analysis required (1) 10 or more left hand–wrist radiographs with a skeletal maturity assessment between chronological ages 3 and 20 years and (2) that the first and last assessments of each radiograph occurred before 5 years and after 13 years of age in girls and before 7 years and after 15 years of age in boys. After these inclusion criteria were applied, the sample included 8463 observations from 345 participants, each with approximately 25 assessments (table 1).

We previously reported a secular trend towards advanced skeletal maturation in FLS participants.¹⁵¹⁶ To ensure that the maturational spurt was not influenced by this secular trend, we assessed maturational trajectories using birth decade as a covariate. No differences were observed in the number of participants exhibiting a maturational spurt across birth decades (data not shown).

Skeletal maturity

Skeletal maturity was assessed using the Fels Hand–Wrist Method,¹⁷ which includes a detailed assessment of up to 98 skeletal indicators in the distal radius, distal ulna and carpus, as well as the metacarpals and phalanges of the first, third and fifth rays. Assessments result in a continuous estimate of skeletal age based on indicator-specific probability density curves derived from the chronological age at which the reference sample exhibited a given maturity indicator (see Roche *et al*¹⁷ for additional details). In the Fels Method, all indicator-specific probability density curves are combined and analysed with a maximum likelihood methodology. This approach limits the influence of a few skeletal indicators that may be substantially advanced and artificially inflating an estimate of skeletal age. Additional details can be found in Roche *et al* and Nahhas *et al*.^{17–19}

Statistical methodology

Statistical modelling and analyses were conducted by sex and between the chronological ages of 3 and 20 years. To characterise and quantify the rate of skeletal maturation throughout childhood and adolescence, we generated three fixed effect statistical models for each participant: (1) linear, (2) fourth-order polynomial and (3) fifth-order polynomial. The underlying longitudinal relationship between chronological age and skeletal age was not known a priori; therefore, maturational trajectories were fit with fixed effect polynomials due to their increased parameter flexibility. Moreover, the use of fourth-order and fifth-order polynomials results in a derivative (ie, maturational tempo) that corresponds, at a minimum, to biologically meaningful maturational milestones (eg, maturational onset, peak and cessation). The childhood phase of growth typically begins at 3 years of age,²⁰ with completion occurring around 6 or 7 years of age in girls and boys, respectively.²¹ Although the majority reach skeletal maturity by 18 years of age,^{11,22} we modelled maturation to 20 years of age to account for participants who may exhibit a delay in the final stages of skeletal maturation. The primary outcome measures of this study are (1) maturational trajectory type (ie, childhood maturational spurt (CMS) only, adolescent maturational spurt (AMS) only or childhood and adolescent maturational spurt (CAMS)); (2) the chronological age at which maturational spurt milestones occur (ie, timing); (3) the rate of maturation at each maturational spurt milestone (ie, tempo); and (4) the duration of time spent between any two maturational milestones, which represents the inverse of longitudinal maturational tempo. All statistical models were generated using the `lm()` function in R 3.4.0.

Linear models

Participant-specific linear models take the following form:

$$Y_{ij} = \beta_{i0} + \beta_{i1}(\text{age}_{i1}) + \epsilon_{ij} \quad (1)$$

where Y_{ij} is the skeletal age of the i th child at the j th time; β_{i0} represents the intercept; β_{i1} represents the slope; and ϵ_{ij} represents error.

Fourth-order and fifth-order fixed effect polynomial models

Participant-specific polynomial models take the following form:

$$Y_{ij} = \beta_{i0} + \beta_{i1}(age_{ij}) + \beta_{i2}(age_{ij})^2 + \dots + \beta_{im}(age_{ij})^m + \epsilon_{ij} \quad (2)$$

where Y_{ij} is the skeletal age of the i th child at the j th time; β_{i0} represents the intercept; and β_{im} represents the polynomial coefficient, where $m = 1, \dots, 4$ or 5 , where 4 or 5 indicates fourth-order or fifth-order polynomials, respectively; and ϵ_{ij} represents error.

Best participant-specific model selection

Model fit for each participant was determined using the Akaike information criterion for finite sample sizes (AICc²³). Models that exhibited the lowest AICc were used for individual maturational trajectories, as well as to estimate the timing and tempo of maturational milestones. The model fit for each participant-specific maturational trajectory was examined by calculating the overall residual SE. Participants whose data fit best with a linear model (38 boys and 28 girls) were classified as linear maturers and were excluded from further analysis.

Maturational milestones and comparisons

Maturational milestones were defined based on the following: (1) childhood peak: maximum velocity of skeletal age proportionate to increases in chronological age between 3 and 6 years of age in girls and 3 and 7 years of age in boys, except in participants categorised as CAMS, where this milestone was defined only as the first observable peak, regardless of chronological age; (2) childhood cessation or adolescent onset: oldest chronological age after the childhood peak, where velocity or acceleration of skeletal age was zero prior to either maturational cessation or the adolescent peak; (3) adolescent peak: maximum velocity of skeletal age proportionate to increases in chronological age after childhood; and (4) adolescent cessation: the youngest chronological age where skeletal age was equal to 18 years and nondecreasing. Maturational milestone estimates derived from fixed effect polynomials that reside near the edge of the observational age range may be influenced by spurious edge effects; therefore, estimates of childhood onset were not considered, and those of adolescent cessation were calculated only from raw observational data. The rate of change in skeletal maturation and the chronological age of attainment for each maturational milestone estimate were determined using the first and second derivatives of the participant-specific best fit model, respectively. The model parameters were as follows:

$$velocity : \frac{Y_{ij}}{\partial(age_{ij})} = \beta_{i1} + 2\beta_{i2} + 3\beta_{i3}(age_{ij})^2 + \dots + m\beta_{im}(age_{ij})^{m-1} \quad (3)$$

$$acceleration : \frac{\partial Y_{ij}}{\partial(age_{ij})^2} = 2\beta_{i2} + 6\beta_{i3}(age_{ij}) + \dots + m(m-1)\beta_{im}(age_{ij})^{m-2} \quad (4)$$

Estimates that fell outside the bounds of the observational data were excluded from further analysis. Differences in maturational timing and tempo for each milestone, as well as the duration of time spent between any two milestones, were compared between maturational trajectory types (ie, CMS, AMS or CAMS) using a one-way analysis of variance.

Mathematical comparisons between identical maturational milestones, identified by the same chronological age criteria (eg, adolescent peak), highlight the underlying biological differences between unique maturational trajectory types. This study was approved by the University of Missouri's Institutional Review Board (project number 2004870, review number 244307).

RESULTS

Identification of a maturational spurt

The relationship between chronological age and skeletal age is non-linear in more than 81% of participants (149 boys and 130 girls), indicating the presence of a maturational spurt in the majority of FLS participants (table 1 and figure 1). Non-linear maturational milestones for 33 participants (12%; 22 boys and 11 girls) were estimated outside the chronological age range of the observational data and were excluded from further analysis.

Three unique non-linear maturational trajectories were identified: CMS, AMS and CAMS. Figure 2 illustrates the fixed effect polynomial and first derivative of a representative participant from each maturational trajectory type. AMS was the most common trajectory type (72%), followed by CAMS (22%) and CMS (6%) in both boys and girls. Longitudinal data were well fit for all participants who exhibited a non-linear relationship between chronological age and skeletal age, with a participant-specific mean residual SE that ranged from 0.245 to 0.379 (figure 3).

Timing of the maturational spurt based on trajectory type

Maturational trajectory type influences the chronological age at which maturational milestones are attained in both boys and girls (table 2 and figure 4). For boys classified as AMS, the mean chronological ages at adolescent onset, peak and cessation were 6.3 (SE 0.17), 12.5 (SE 0.22) and 17.5 (SE 0.08) years, respectively. The mean estimates for adolescent onset and peak were 3.6 ($p<0.001$) and 2.0 ($p<0.001$) years later in boys classified as CAMS when compared with those classified as AMS. The chronological ages at which childhood peak was attained were similar ($p = 0.9$) for boys classified as either CMS (5.6, SE 0.22) or CAMS (5.6, SE 0.15); however, the chronological age at which childhood cessation or adolescent onset was attained was significantly later ($p<0.001$) in boys classified as CMS (13.5, SE 0.51) when compared with CAMS (9.9 SE 0.20). In girls, similar trends were observed.

Tempo of the maturational spurt based on trajectory type

The tempo of skeletal maturation (ie, rate of change in skeletal age proportionate to chronological age) differed among trajectory types in both boys and girls (table 2 and figure 4). In boys classified as AMS, the rates of skeletal maturation occurring at adolescent onset and peak were 0.8 (SE 0.02) and 1.3 (SE 0.02), respectively. Maturational rate differed among trajectory types in boys only at the childhood cessation or adolescent onset milestone ($p=0.002$). In girls, differences were observed only for milestone estimates of childhood peak ($p=0.04$) between those classified as either CMS or CAMS.

Duration of the maturational spurt based on trajectory type

The duration of time spent between any two maturational milestones is significantly different among trajectory types (table 2 and figure 4), with participants classified as CAMS spending less time between nearly all maturational milestones than when compared with either CMS or AMS. In boys, the largest difference ($p < 0.001$) occurred between adolescent onset and adolescent cessation for those classified as AMS (11.2, SE 0.20) and CAMS (7.3, SE 0.28). Similar estimates were observed in girls.

DISCUSSION

Historically, the relationship between skeletal age and chronological age throughout childhood and adolescence has been described by a linear paradigm, where ‘average’ maturing children exhibit a 1:1 relationship between skeletal age and chronological age. However, longitudinal assessments of skeletal maturation in a single individual often progress in a non-linear fashion, particularly during adolescence. Despite this observation, deviations in the tempo of skeletal maturation have been attributed to normal stochastic variation and remain largely uncharacterised. This study challenged this paradigm and found that less than 20% of participants exhibited a linear relationship between chronological age and skeletal age. The majority of participants were characterised by one of three non-linear maturational trajectories (eg, CMS, AMS and CAMS), suggesting that the relationship between chronological age and skeletal age is non-linear.

The primary limitation of this study is the homogeneity of the FLS participants in regard to their racial and/or ethnic diversity. Known differences in the timing of skeletal maturation among groups may limit the generalisability of the chronological age at which maturational milestones are attained^{24–27}; the non-linear nature of skeletal maturation, however, has also been suggested in other racial groups,^{28,29} indicating the likelihood of the presence of a maturational spurt.

The ability to anticipate a maturational spurt in a variety of paediatric settings may significantly influence patient prognosis, particularly when treatment type, timing and management are directly related to biological maturity status and the attainment of specific growth and/or developmental milestones. Although non-linear changes in the tempo of skeletal maturation as a whole have been noted in several populations using distinct skeletal maturity assessment methods,^{1,2,28,29} this is, to our knowledge, the first time that maturational milestones associated with participant-specific maturational trajectories have been characterised. Previous indications of rapid maturation in other populations correspond to the AMS we identified here. For example, the chronological age at which the adolescent peak in boys classified as AMS is attained resembles the single period of rapid maturation identified by Loesch and colleagues.²⁸ However, estimate differences in girls classified as AMS were approximately 2 years later than when compared with the estimates provided by Loesch *et al.*² More recently, Cole *et al.*²⁹ identified two distinct periods of rapid maturation between 9 and 20 years of age that are similar to the individual mean estimates we observed for the adolescent peak milestone in participants classified as either AMS or CAMS. In both Loesch *et al.*²⁸ and Cole *et al.*²⁹ observational data during childhood were not reported, making it difficult to detect underlying differences in maturational trajectory shape that

began during childhood and thus its influence on milestone variation. The reported differences and potential bias in the timing of the maturational spurt could be influenced by population-specific differences in skeletal maturation, skeletal maturity assessment methods or mathematical modelling techniques. However, the underlying presence of a maturational spurt appears to be consistent across these populations.

It is well known that biological sex plays a critical role in the timing of growth and/or developmental milestones (eg, development of axillary hair or peak height velocity). It is, therefore, not surprising that the chronological age of attainment and the rate of skeletal maturation that occur at maturational milestones are different between boys and girls, especially when accounting for maturational trajectory type. Yet, the mathematical differences between milestone estimates for boys and girls are smaller than we would have expected given other developmental milestones, including those surrounding adolescence. If sex-specific patterns of androgen release played an integral part in the presence of a maturational spurt, as it does in many other adolescent milestones, we would expect girls to attain these milestones at much earlier chronological ages than in boys. Moreover, at this chronological age, girls tend to exhibit more mature developmental phenotypes than when compared with boys,³⁰³¹ suggesting that the skeletal phenotypes driving the maturational spurt are likely unique among boys and girls.

In conclusion, we have identified and characterised nonlinear changes in the tempo of skeletal maturation throughout childhood and adolescence. Although the maturational paradigm suggests that average skeletal maturation progresses linearly, our work provides evidence that, in fact, skeletal maturation progresses in a non-linear fashion. Moreover, changes in the tempo of skeletal maturation are patterned and represent more than just random interindividual variability. This work represents a critical shift in the maturational paradigm, particularly in how we interpret deviations in the relationship between chronological age and skeletal age during normal growth and development and their relationship to other biological milestones (eg, peak height velocity).

Acknowledgements

We thank the participants in the Fels Longitudinal Study for their commitment to research. Thank you to Kimberly Lever, Sharon Lawrence, Carol Cottom, Beverly Barry, Nicole Odom and Christina Holzhauser for their skeletal maturity assessment expertise, as well as Kevin Middleton for his statistical guidance. The content represented herein is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Funding This work was supported by funds from the National Institutes of Health (NIH, F31 HD091939 and R01 AR055927). Participant radiographs were collected previously with research funds also provided by the NIH (R01 HD012252).

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What is already known on this topic?

- ▶ Skeletal maturation plays a critical role in understanding growth and developmental processes, including interpretations of biologically relevant milestones (eg, peak height velocity).
- ▶ The current paradigm for the tempo of skeletal maturation suggests that skeletal age tracks linearly with chronological age.
- ▶ Maturational deviations from linearity are often attributed to either environmental (eg, nutrition) or genetic factors, but not to normal fluctuations in tempo.

What this study adds?

- ▶ We identified the presence of a maturational spurt during normal childhood and/or adolescence using participant-specific maturational trajectories.
- ▶ Maturational trajectories and milestone estimates (eg, age at peak maturational velocity) are now defined.
- ▶ These data provide evidence for a shift in the paradigm of skeletal maturation from linear to non-linear, particularly during adolescence.

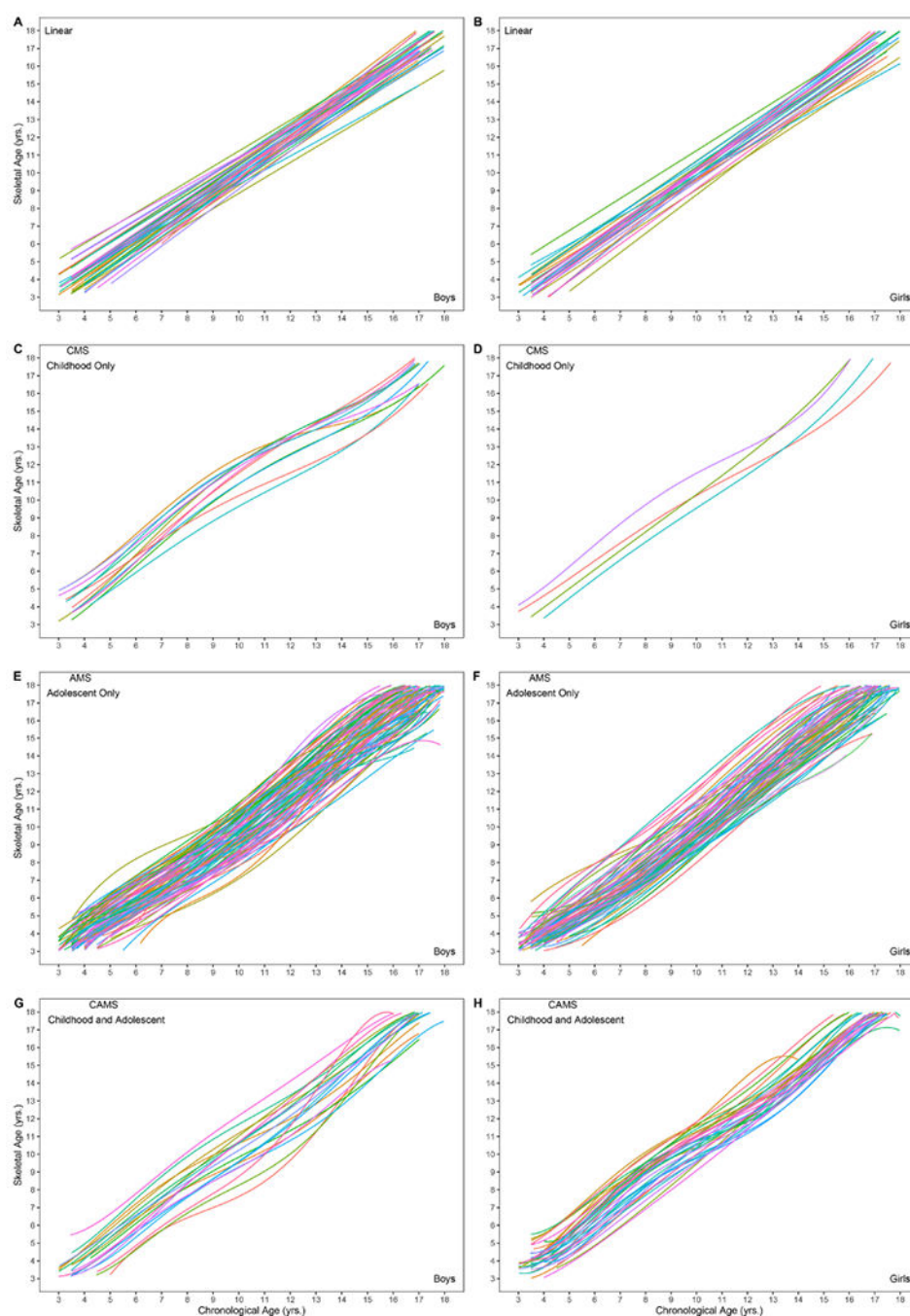


Figure 1.

Individual maturational trajectories: each maturational trajectory type is depicted separately, with each individual represented by a unique colour for linear maturers (A,B), CMS (C,D), AMS (E,F) and CAMS (G,H) in both boys (right panel; A,C,E,G) and girls (left panel; B,D,F,H). This image is the property of the authors. AMS, adolescent maturational spurt; CAMS, childhood and adolescent maturational spurt; CMS, childhood maturational spurt.

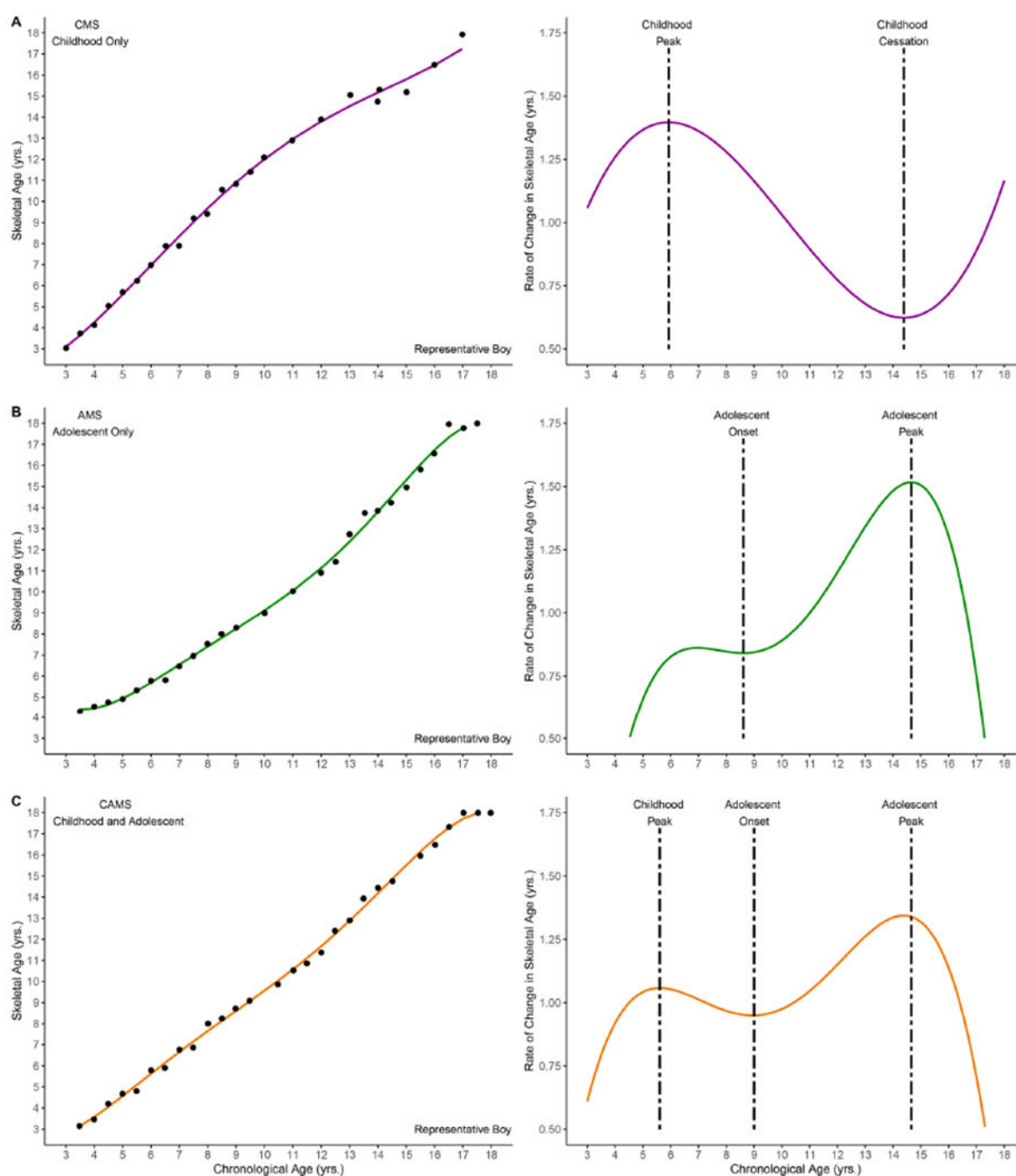


Figure 2.

Representative polynomial curve and first derivative: individual (dots) observations, polynomial model fit (solid line, right panel) and first derivative with all relevant maturational milestones illustrated (dotted line, left panel) for a representative participant classified as CMS(A, illustrated in purple), AMS (B, illustrated in green), and CAMS (C, illustrated in orange). This image is the property of the authors. AMS, adolescent maturational spurt; CAMS, childhood and adolescent maturational spurt; CMS, childhood maturational spurt.

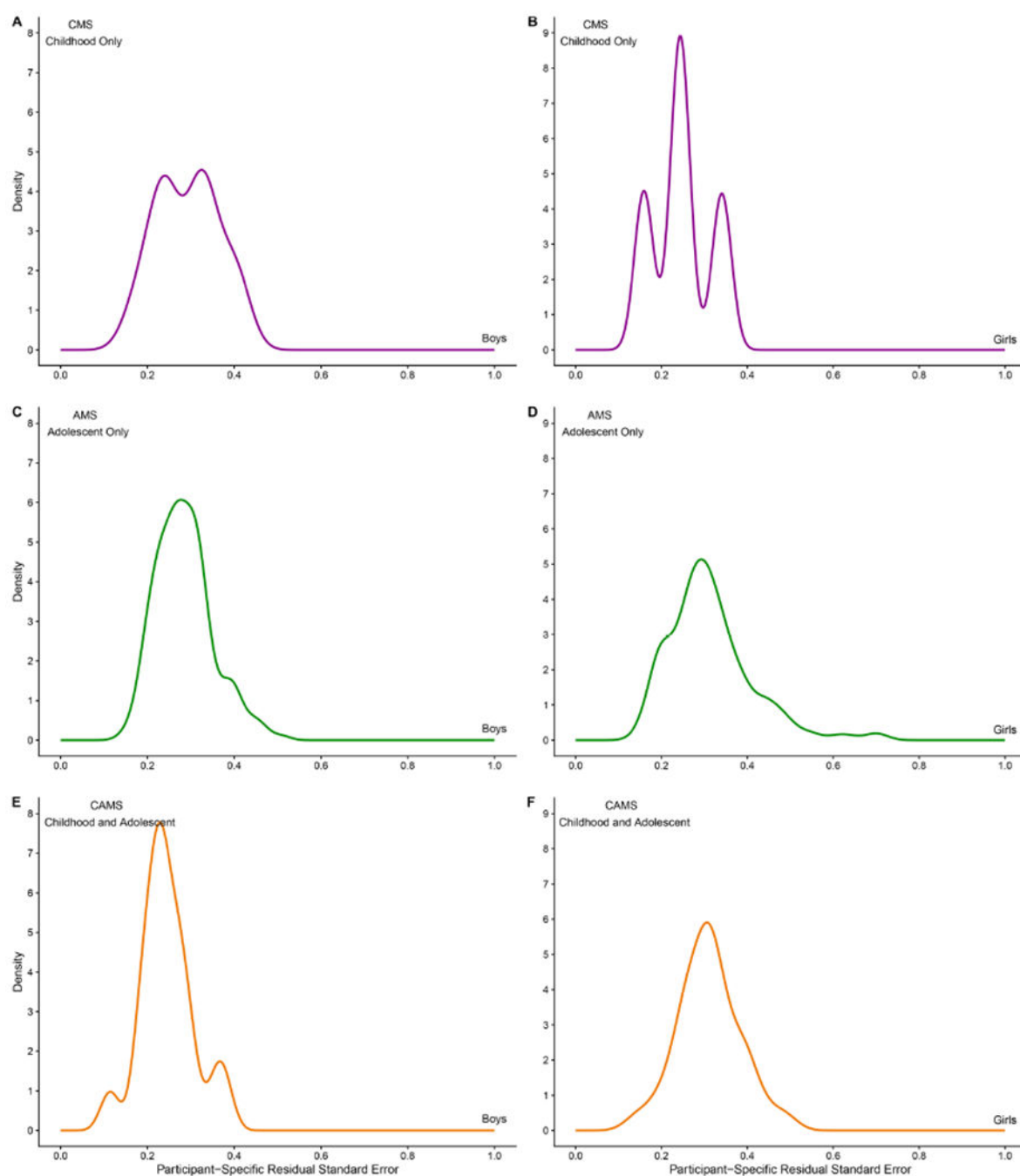


Figure 3.

Participant-specific residual SE for model fit: density plots depicting the residual SE of participant-specific model fit for those classified as CMS (A,B; illustrated in purple), AMS (C,D; illustrated in green), and CAMS (E,F; illustrated in orange) in boys (right panel; A,C,E) and girls (left panel: B,D,F). This image is the property of the authors. AMS, adolescent maturational spurt; CAMS, childhood and adolescent maturational spurt; CMS, childhood maturational spurt.

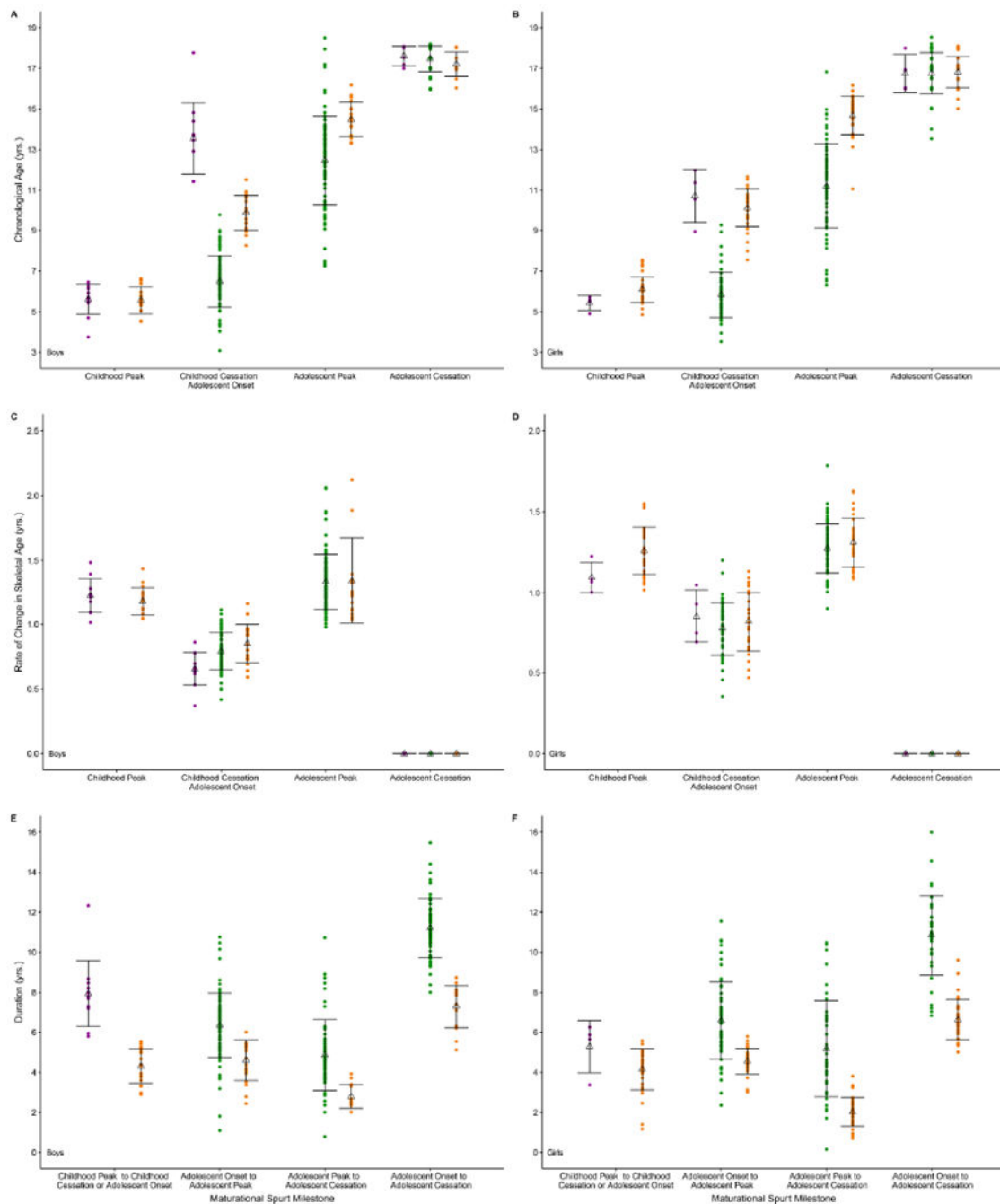


Figure 4.

Timing, tempo and duration estimates of the maturational spurt: individual (dots) and trajectory-type mean (represented by a P) with SD bars for each maturational spurt milestone estimate of chronological age (top panel; A,B), rate (middle panel; C,D) and duration (bottom panel; E,F) in both boys (right panel; A, C, and E) and girls (left panel; B, D, and F), where purple indicates participants classified as CMS; green indicates those classified as AMS; and orange indicates CAMS. This image is the property of the authors.

AMS, adolescent maturational spurt; CAMS, childhood and adolescent maturational spurt;
CMS, childhood maturational spurt.

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Table 1

Sample statistics for all participants who met the inclusion criteria, represented by either the range or the mean (median, IQR)

Sex	Characteristic	All participants	Linear	Excluded Non-linear	Trajectory type		
					CMS	AMS	CAMS
Boys	Total participants	187	38	22	12	96	19
	Total observations	4603	868	678	294	2393	477
	Age range (years)	3.0–19.2	3.0–18.5	3.0–18.2	3.0–18.5	3.0–19.2	3.0–18.3
	Skeletal age range (years)	1.1–18.0	1.9–18.0	2.25–18.0	2.7–18.0	1.1–18.1	1.65–18.0
	Mean observations per participant	25.3 (26.0, 6.0)	22.8 (23.0, 5.8)	26.0 (26.5, 5.5)	24.5 (23.5, 5.5)	24.9 (25.5, 6.0)	25.1 (26.0, 4.5)
Girls	Total participants	158	28	11	4	81	34
	Total observations	3860	678	285	85	1935	877
	Age range (years)	3.0–20.0	3.0–18.5	3.0–20.0	3.0–18.0	3.0–18.5	3.0–18.1
	Skeletal age range (years)	1.7–18.0	1.9–18.0	2.4–18.0	2.9–18.0	1.7–18.0	2.48–18.0
	Mean observations per participant	24.8 (25.0, 5.0)	24.2 (24.0, 5.0)	25.9 (27.0, 3.5)	21.2 (21.0, 0.8)	23.9 (23.0, 4.0)	25.8 (26.0, 3.8)

Participants were further divided by maturational trajectory type, including those with non-linear skeletal maturation that exhibited milestone estimates outside the observational age range. AMS, adolescent maturational spurt; CAMS, childhood and adolescent maturational spurt; CMS, childhood maturational spurt.

Table 2

Maturational spurt milestone estimates by trajectory type: mean chronological age of milestone attainment, mean rate (ie, tempo) of maturation at milestone attainment, duration of time spent between each milestone and the SE of each milestone estimate.

Sex	Estimate	Milestone	Trajectory type				ANOVA
			CMS	AMS	CAMS		
Boys	Mean chronological age (years)	Childhood peak	5.6 (0.22)	–	5.6 (0.15)	=0.9	
		Childhood cessation/adolescent onset	13.5 (0.51)	6.3 (0.17)	9.9 (0.20)	<0.001	
		Adolescent peak	–	12.5 (0.22)	14.5 (0.19)	<0.001	
		Adolescent cessation	17.6 (0.17)	17.5 (0.08)	17.2 (0.16)	=0.2	
	Mean rate (change in skeletal age per year)	Childhood peak	1.2 (0.04)	–	1.2 (0.02)	=0.3	
		Childhood cessation/adolescent onset	0.7 (0.04)	0.8 (0.02)	0.9 (0.03)	=0.002	
		Adolescent peak	–	1.3 (0.02)	1.3 (0.08)	=0.9	
	Mean duration (years)	Childhood peak to cessation/adolescent onset	7.9 (0.48)	–	4.3 (0.20)	<0.001	
		Adolescent onset to adolescent peak	–	6.4 (0.17)	4.6 (0.23)	<0.001	
		Adolescent peak to adolescent cessation	–	4.9 (0.23)	2.8 (0.15)	<0.001	
		Adolescent onset to adolescent cessation	–	11.2 (0.20)	7.3 (0.28)	<0.001	
Girls	Mean chronological age (years)	Childhood peak	5.4 (0.18)	–	6.1 (0.12)	=0.05	
		Childhood cessation/adolescent onset	10.7 (0.66)	5.3 (0.25)	10.1 (0.16)	<0.001	
		Adolescent peak	–	11.2 (0.23)	14.7 (0.16)	<0.001	
		Adolescent cessation	16.7 (0.47)	16.8 (0.14)	16.8 (0.13)	=1.0	
	Mean rate (change in skeletal age per year)	Childhood peak	1.1 (0.05)	–	1.3 (0.03)	=0.04	
		Childhood cessation/adolescent onset	0.9 (0.08)	0.8 (0.02)	0.8 (0.03)	=0.4	
		Adolescent peak	–	1.3 (0.02)	1.3 (0.03)	=0.2	
	Mean duration (years)	Childhood peak to cessation/adolescent onset	5.3 (0.65)	–	4.0 (0.21)	=0.06	
		Adolescent onset to adolescent peak	–	6.6 (0.26)	4.6 (0.11)	<0.001	
		Adolescent peak to adolescent cessation	–	5.2 (0.36)	2.0 (0.12)	<0.001	
		Adolescent onset to adolescent cessation	–	11.0 (0.36)	6.6 (0.17)	<0.001	

P values are rounded to the nearest significant digit of precision as outlined by Cole.³²

AMS, adolescent maturational spurt; ANOVA, analysis of variance; CAMS, childhood and adolescent maturational spurt; CMS, childhood maturational spurt.